

Figure 1 — EnvMeta Five-Tier Framework Architecture

Streamlit GUI: upload → auto-recognize 11 file types → analyze → export

L1 General inference engine

S1 CLR debias + S2 999-perm null + S3 MCDA + Bradford-Hill veto + Saltelli $\pm 20\%$

L2 Six-claim YAML hypothesis schema

pathway_active / inactive / coupling / env_correlation / keystone / group_contrast

L3 Plugin framework (post-acceptance)

user-supplied Python analyze() modules auto-register in GUI

L4 Fork Bundle distribution

Bundle.zip = KB + YAML + config + KEGG snapshot + sample data

L5 KEGG-driven biogeochemical KB

4 elements $\times$ 18 pathways $\times$ 57 KO targets (KB v2.0)

data
flow

